Gene content MDS · 52 manuscript strains

Classical MDS on (1 - Jaccard) distance of cluster presence/absence

G. thermodenitrificans NG80-2
G. thermodenitrificans KCTC3902
G. thermodenitrificans T12
G. thermodenitrificans ID-1
G. thermodenitrificans K1041
Geobacillus sp. 47C-IIb

P. caldoxylosilyticus KH3-5
P. caldoxylosilyticus KH1-6
P. caldoxylosilyticus KH1-5
P. caldoxylosilyticus ER4B

G. subterraneus OXR-76
Geobacillus sp. 46C-IIa
Geobacillus genomosp. 3
G. subterraneus KCTC3922
G. subterraneus CPW16

P. toebii NEB718
P. toebii NBRC 107807
Parageobacillus sp. KH3-4

P. thermoglucosidasius NCIMB 11955
P. thermoglucosidasius TM242
P. thermoglucosidasius DSM 2542
P. thermoglucosidasius BH4-1
P. thermoglucosidasius C56-YS93

G. stearothermophilus SJEF4-2
G. stearothermophilus EF60063
G. stearothermophilus B5
G. stearothermophilus ATCC 12980
G. subterraneus E55-1
G. thermoleovorans FJAT-2391
G. stearothermophilus DSM 458
G. kaustophilus T-1
Geobacillus sp. GHH01
Geobacillus sp. JS12
G. zalihae JS1-2
Geobacillus sp. Y412MC52
G. zalihae SURF-189
G. thermoleovorans KCTC 3570
G. thermocatenulatus KCTC 3921
G. stearothermophilus DG-1
G. thermoleovorans V0
G. stearothermophilus EF60133
G. stearothermophilus EF60045
Geobacillus sp. C56-T3
G. zalihae SURF-114
G. stearothermophilus EF60134
G. kaustophilus HTA426
G. caldolyticus NEB414
G. proteiniphilus T-6
G. thermoleovorans SURF-48B
G. thermoleovorans ARTRW1
G. thermoleovorans CCB_US3_UF5
G. kaustophilus NBRC 102445

0.2
0.0
-0.2
-0.50
-0.25
0.00
0.25
Dim 2 (10.7%)
Dim 1 (31.4%)

Species labels (Figure 2)

G. caldolyticus · *G. genomosp.* · *G. kaustophilus* · *G. proteiniphilus* · sp. (*G./P.*) · *G. stearothermophilus* · *G. subterraneus*
G. thermocatenulatus · *G. thermodenitrificans* · *G. thermoleovorans* · *G. zalihae* · *P. caldoxylosilyticus* · *P. thermoglucosidasius* · *P. toebii*

All 14,337 pan-genome clusters. Point coordinates and species colors are unchanged.